



Mark Scheme (Results)

June 2025

Pearson BTEC Nationals in
Applied Science/Forensic and Criminal
Investigation (31617H/1P)
Unit 1: Principles and Applications of Science I
(Physics)

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Unit 1: Principles and Applications of Science I (Physics)

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1(a)	transverse	ignore progressive ignore stationary	1
1(b)(i)	C arrow Y		1
1(b)(ii)	A arrow W		1
Total = 3 marks			

Question Number	Answer	Additional Guidance	Mark
2(a)	digital	accept binary	1
2(b)	any two from: no waves transmitted at wavelength of 0.5 cm (1) percentage (of waves passing through the atmosphere) increases as wavelength increases (between 0.5 and 0.9 cm) (1) non-linear (1)	accept plateaus accept rate of increase slows down (after 0.6 cm)	2
2(c)	reflected by the <u>ionosphere</u> (1) by {electrons / ions / charged particles} (1)	accept absorbed in the ionosphere	2
Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
3(a)	ray refracts (1) AND any one of: towards the normal (1) slows down (1)	accept changes direction accept bends ignore references to total internal reflection accept ray of light drawn on the diagram showing refraction towards the normal (2)	2
3(b)	substitution (1) $1.6 = \frac{3.0 \times 10^8}{v}$ rearrangement (1) $(v =) \frac{3.0 \times 10^8}{1.6}$ evaluation (1) $1.9 \times 10^8 \text{ (m s}^{-1}\text{)}$	an answer of $1.9 \times 10^8 \text{ (m s}^{-1}\text{)}$ rounded correctly with or without workings scores 3 marks accept substitution and rearrangement in either order accept $(v =) \frac{c}{n}$ accept answers in non-standard form	3

3(c)	ray of light travels (through the endoscope / optical fibres) by total internal reflection (1) ray of light travels through one <u>bundle</u> of optical fibres to (illuminate) {stomach / lining} (1) ray of light is reflected (from the lining) into a second <u>bundle</u> (of optical fibres towards the eyepiece) (1) each fibre gives a small part of the complete image (1)	accept TIR reject refraction reject refraction accept fibres must be in coherent bundles (to allow image formation) accept correct references to pixels	4
Total = 9 marks			

Question Number	Answer	Additional Guidance	Mark
4(a)	sensor A is closer (to the transmitter) than sensor B (1) {energy / signal} (from the transmitter) is spread out {more / over a larger area} (as distance increases) (1)	ORA accept correct reference to inverse square law	2
4(b)	measurement of distance to sensors A and B (1) (0.4 – 0.1 =) 0.3 (m) AND (1.3 – 0.1 =) 1.2 (m) substitution (1) $\frac{5.0}{(I_B)} = \frac{(1.2)^2}{(0.3)^2}$ rearrangement (1) $(I_B) = \frac{5.0}{1.2^2} \times 0.3^2$ evaluation (1) 0.31 (units)	an answer of 0.3125 rounded correctly with or without workings scores 4 marks ecf throughout accept substitution and rearrangement in either order answers based on distances of 0.4 and / or 1.3 score a maximum of 3 marks	4
Total = 6 marks			

Question Number	Answer	Additional Guidance	Mark
5(a)	D mass per unit length		1

Question Number	Indicative Content
5(b)	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content but learners should be rewarded for other relevant answers.</i> <u>strings</u> • friction between bow and string • transfers energy to the string • strings vibrate • a stationary / standing wave is set up on the string • this causes energy to be transferred to the body of the violin • pitch of sound related to string properties <u>violin</u> • the body of the violin resonates • resonates at the same frequency as the string • f shaped holes amplify sound <u>wave</u> • the strings / instrument vibrating causes longitudinal / pressure waves to be set up in the air • which can be heard as sound • sound waves are longitudinal • which consist of compressions and rarefactions • stationary / standing waves are caused by waves reflecting from each end of the string • the reflected waves superpose • the superposed waves form antinodes • by constructive interference • in phase • and nodes • by destructive interference • out of phase <u>harmonics</u> • fundamental frequency is called the first harmonic • relation of harmonics to wavelength/frequency

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge of scientific facts / concepts with generalised comments made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The discussion shows some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific knowledge facts / concepts to provide the discussion being presented. • Lines of argument mostly supported through the application of relevant evidence. • The discussion shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts / concepts to provide the discussion being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The discussion shows a well-developed structure which is clear, coherent and logical.
Total = 7 marks		

